RACS — Final Project Overview

Retrieval-Augmented Coverless Semantic Steganography

> ***Purpose:*** *Compact, complete project specification for an LLM or developer. It defines the original AgentSemSteCom system, the RACS modification, architecture, token/security design, implementation boundaries, evaluation, risks, hypotheses, and success criteria. It intentionally removes repetition from the longer overview.*

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1. Project Definition

RACS (Retrieval-Augmented Coverless Semantic Steganography) starts from the 2026 IEEE paper **"Secure Intellicise Wireless Network: Agentic AI for Coverless Semantic Steganography Communication."**

The paper's **AgentSemSteCom** combines:

- agentic semantic extraction,
- implicit structural features,
- diffusion-based reference-image generation,
- EDICT-based reversible diffusion,
- digital-token security,
- semantic communication,
- wireless transmission,
- optional task-oriented enhancement.

Problem

The original system generates a reference image through an expensive diffusion process. RACS investigates:

> ***Can retrieval-based reference-image acquisition provide sufficiently compatible semantic and structural conditioning for reversible coverless steganography while reducing the computational cost of reference generation?***

RACS does **not** assume retrieval is better. It is a quality-security-robustness-efficiency comparison.

Core modification

```
AgentSemSteCom:
Secret → Kpub + Xfeat → diffusion generation → Xref

RACS:
Secret → Kpub + Xfeat → retrieval → reranking
      → token-controlled selection → optional light editing → Xref
```

The downstream EDICT, token perturbation, semantic communication, channel and recovery pipeline should initially remain unchanged.

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2. Original AgentSemSteCom

2.1 End-to-end flow

```
Secret Xs
  ↓
Agentic semantic analysis
  ↓
Kpub + Xfeat
  ↓
Diffusion/DDIM reference generation
  ↓
Xref
  ↓
EDICT + token perturbation
  ↓
Xstego
  ↓
Semantic communication
  ↓
Wireless channel
  ↓
Semantic decoder
  ↓
Reconstructed stego
  ↓
EDICT inversion + inverse token perturbation
  ↓
Recovered secret
```

2.2 Stage 1 — Semantic extraction

Public semantic key:

```
Kpub = fLLM(Xs)
```

`Kpub` describes what the secret image contains.

Implicit structural feature:

```
Xfeat = G(Xs, R)
```

`Xfeat` describes structural/layout information. The paper discusses:

- semantic segmentation for common images,
- OpenPose skeletal landmarks for facial images,

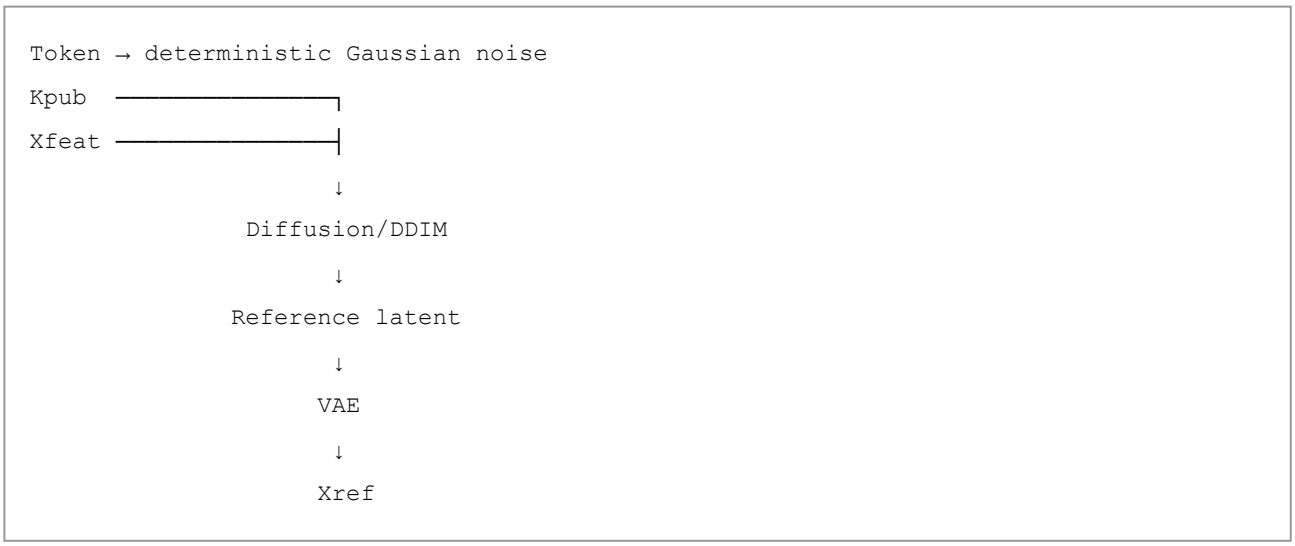
- style-related conditions where explicit reference features are not required.

Core distinction:

```
Kpub  = semantic/content information
Xfeat = structural/layout information
```

2.3 Stage 2 — Reference generation

The original reference generator uses diffusion/DDIM, structural conditioning such as ControlNet, IPAdapter-plus, and the digital token.



This is the main computational stage RACS replaces.

2.4 Stage 3 — Coverless steganography

Reported components:

- Stable Diffusion v1.5,
- EDICT,
- coupled latent trajectories,
- token-controlled latent perturbation,
- VAE.

Conceptually:

```
Secret
 ↓
VAE encoder
 ↓
Secret latent
 ↓
Forward diffusion
 ↓
Noisy latent
 ↓
Token-controlled perturbation
 ↓
EDICT reverse process conditioned by Kpub + Xref
 ↓
VAE decoder
 ↓
Stego image
```

EDICT is important because the process is intended to be reversible/invertible for secret recovery.

2.5 Digital token

The original token has two roles:

```
Token
 ├─→ reference-generation seed
 └─→ latent perturbation/security control
```

The token therefore provides dual-stage protection.

2.6 Stage 4 — Semantic communication

Analog setup:

- SwinJSCC,
- AWGN,
- SNR 5/10/15/20 dB,
- `h = 1`,
- `C = 48`,
- training SNR = 10 dB.

Digital setup:

- MDJCM,
- DIV2K,
- modulation order 64,
- SNR 10 dB.

2.7 Stage 5 — Receiver

```
Received representation
 ↓
Semantic decoder
 ↓
Reconstructed stego
 ↓
Reference conditions
 ↓
EDICT inversion
 ↓
Inverse token perturbation
 ↓
Recovered secret
```

2.8 Optional enhancement

The paper uses CodeFormer for optional facial enhancement:

- FFHQ,
- 512×512,
- fidelity weight 0.5.

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3. Original Parameters and Results

Component	Reported configuration
Diffusion	Stable Diffusion v1.5
EDICT steps	50
Perturbation strength η	0.05
EDICT mixing coefficient	0.93
Edit strength	0.5

Component	Reported configuration
Reference guidance	IPAdapter-plus
IPAdapter weight	1
Analog SemCom	SwinJSCC
Digital SemCom	MDJCM
Channel	AWGN
SNR	5, 10, 15, 20 dB
Channel coefficient	h = 1
Analog bandwidth	C = 48
Digital modulation	64
Facial enhancement	CodeFormer
CodeFormer fidelity	0.5

Metrics

Metric	Direction
PSNR	Higher
SSIM	Higher
MSE	Lower
LPIPS	Lower

Reported reference values

Facial, 5 dB:

```
PSNR = 22.73 dB
SSIM = 0.6989
```

Facial, 20 dB:

```
PSNR = 26.29 dB
LPIPS = 0.2127
```

Style, 5 dB:

```
MSE   = 0.721 × 10^-2
LPIPS = 0.3850
```

AgentSemSteCom vs SemSteDiff, common images, 5 dB:

```
PSNR: 19.71 vs 18.04 dB
MSE:  1.207×10^-2 vs 1.73×10^-2
```

At 10 dB:

```
SSIM: 0.5643 vs 0.5183
```

CodeFormer, facial, 5 dB:

```
             Before     After
PSNR         22.74      23.15
SSIM          0.6989      0.7244
MSE           0.566e-2    0.514e-2
LPIPS         0.4042      0.2928
```

At 20 dB, enhanced PSNR is 25.59 dB versus baseline 26.29 dB, demonstrating why multiple metrics are necessary.

Security baseline

```
Correct token → PSNR 25.37 dB
Wrong token   → PSNR 18.91 dB
Security gap  → 6.46 dB
```

---

4. RACS Architecture

4.1 Main pipeline

```
Secret Xs
  ↓
Semantic extraction → Kpub
Structure extraction → Xfeat
  ↓
Retrieval query
  ↓
CLIP/VLM embedding
  ↓
FAISS image index
  ↓
Top-K candidates
  ↓
Semantic + structural reranking
  ↓
High-quality candidate pool
  ↓
Digital token
  ↓
Deterministic candidate selection
  ↓
Optional lightweight editing
  ↓
Xref
  ↓
Existing EDICT + token perturbation
  ↓
Xstego
  ↓
Existing semantic communication
  ↓
Wireless channel
```

4.2 Receiver

```
Received semantic representation
  ↓
Semantic decoder
  ↓
Reconstructed stego
  ↓
Same query/configuration
  ↓
Same database + embedding + FAISS
  ↓
Same ranking
  ↓
Correct token
  ↓
Same Xref
  ↓
EDICT inversion
  ↓
Inverse token perturbation
  ↓
Secret
```

---

5. Retrieval Database

The research plan proposes a manageable **LAION subset**, initially around **50k–100k images**, with smaller subsets appropriate for development.

Each image should have:

```
image_id
image
CLIP/VLM embedding
semantic metadata/caption if available
structural representation
optional steganographic-suitability features
```

Indexing is offline:

```
Corpus → embedding model → embeddings → FAISS index
```

Online processing should mainly be:

```
query embedding
+ nearest-neighbour search
+ reranking
+ optional editing
```

This separation is essential to the efficiency argument.

---

6. Retrieval and Reranking

6.1 First retrieval layer

```
Secret
 ↓
Kpub / semantic description
 ↓
CLIP text or multimodal embedding
 ↓
FAISS
 ↓
Top-K
```

CLIP provides semantic similarity but does **not** guarantee:

- same pose,
- same geometry,
- same layout,
- same object arrangement,
- same texture,
- steganographic suitability.

Therefore CLIP-only retrieval is the baseline, not the final design.

6.2 Structural reranking

Candidate score:

```
Score(I) =
    α × Ssemantic(I)
  + β × Sstructural(I)
```

Possible extension:

```
Score(I) =
    α Ssemantic
  + β Sstructural
  + γ Ssteganographic
```

Structural representations may include:

- segmentation,
- edges/layout,
- depth,
- object arrangement,
- pose landmarks,
- facial landmarks.

The original paper's structural-feature results motivate this component.

---

7. Digital Token in RACS

The token should be retained but its reference-acquisition role changes.

Original:

```
Token → diffusion noise seed
```

RACS:

```
Top-K candidates
      ↓
Token
      ↓
deterministic mapping
      ↓
selected candidate
```

Example conceptual mapping:

```
index = H(token) mod K
```

The exact mapping must be fixed.

Quality constraint

Never use:

```
Token → arbitrary database image
```

Use:

```
Query
 ↓
Top-K relevant candidates
 ↓
quality filtering/reranking
 ↓
valid candidate pool
 ↓
token selection
```

The token provides controlled diversity/security while preserving candidate quality.

Security hypothesis

If the valid candidate pool has size K:

```
P(correct reference | no token) ≈ 1/K
```

and a simple uncertainty measure is:

```
log2(K) bits
```

This is only a research model, not a complete security proof. A full analysis must consider token entropy, database knowledge, ranking knowledge, candidate filtering, side information, and attacker capabilities.

---

8. VLM / Agentic Reranking

A stronger RACS variant can use a VLM/agent after FAISS:

```
FAISS
 ↓
Top-K
 ↓
structural filtering
 ↓
VLM/agent evaluation
 ↓
fixed ranking
 ↓
token-controlled selection
```

The VLM can consider:

- semantic compatibility,
- structure/layout,
- visual similarity,
- editing requirement,
- possible steganographic suitability.

Important protocol rule

Do not put a nondeterministic cloud LLM directly in the critical transmitter/receiver selection path.

Prefer:

```
VLM → candidate scores
     ↓
fixed deterministic ranking
     ↓
token → deterministic selection
```

using a shared/frozen/local model or deterministic scoring process.

Compare:

```
RACS-Base
vs
RACS-Agent
```

to determine whether the added intelligence justifies its cost.

---

9. Optional Lightweight Editing

Retrieval may find an image that is semantically good but structurally imperfect.

```
Retrieved image + Xfeat
        ↓
light image-to-image / structure-guided editing
        ↓
improved Xref
```

Possible directions:

- SDEdit-style partial diffusion,
- lightweight image-to-image adjustment,
- structure-guided editing.

The editing stage must remain substantially cheaper than full reference generation.

Compare:

```
Retrieval
vs
Retrieval + editing
vs
Full generation
```

on both quality and latency.

---

10. Optional Retrieve-or-Generate

RACS can retain diffusion as a fallback:

```
Secret
 ↓
Retrieve
 ↓
Candidate quality
 ├─ good → retrieve
 └─ poor → diffusion generation
```

Potential systems:

```
Always Generate
Always Retrieve
Retrieve + Edit
Retrieve-or-Generate
```

The hybrid architecture is useful because it applies expensive generation only when retrieval is inadequate.

---

11. Scientific Comparison Boundary

The first RACS comparison should change **only reference acquisition**.

Keep:

```
Secret images       → same
Kpub extraction     → same
Xfeat extraction    → same
EDICT               → same
Token perturbation  → same
Semantic Comms      → same
Channel             → same
Recovery            → same
Metrics             → same
```

Change:

```
Reference acquisition:
diffusion generation → retrieval
```

This is essential for attribution.

---

12. Research Hypotheses

H1 — Efficiency

Retrieval reduces:

- reference latency,
- GPU computation,
- memory,
- potentially energy.

H2 — Quality

A sufficiently compatible retrieved reference, optionally edited, can maintain acceptable:

- PSNR,
- SSIM,
- MSE,
- LPIPS.

H3 — Security

Token-controlled selection can preserve a meaningful correct-token/wrong-token recovery gap.

H4 — Robustness

RACS remains usable at:

```
5, 10, 15, 20 dB
```

even if performance differs from AgentSemSteCom.

---

13. Evaluation Framework

Systems

At minimum:

```
1. AgentSemSteCom
2. RACS Retrieval-only
3. RACS Retrieval + Light Editing
4. RACS Retrieve-or-Generate
```

Optional intelligent variant:

```
5. RACS + VLM/Agent Reranking
```

Image classes

Use the baseline categories where applicable:

```
common
facial
style
```

SNR

```
5 dB
10 dB
15 dB
20 dB
```

Quality

```
PSNR ↑
SSIM ↑
MSE ↓
LPIPS ↓
```

---

14. Efficiency Evaluation

This is the major new evaluation dimension.

Measure:

Reference acquisition

```
T_generation
T_retrieval
T_reranking
T_editing
```

End-to-end latency

```
semantic extraction
+ reference acquisition
+ steganography
+ communication
+ recovery
```

Memory

Peak GPU memory for:

- generation,
- retrieval,
- VLM/reranking,
- editing,
- full pipeline.

Computation

Where feasible:

- FLOPs,
- diffusion steps,
- model forward passes.

Optional:

- energy,
- CPU utilization,
- GPU utilization.

Do not claim RACS is faster until measured.

---

15. Security and Steganalysis

Security

At minimum:

```
Correct token
Wrong token
No token
```

Measure:

```
PSNR
SSIM
MSE
LPIPS
```

Compare against the original security baseline:

```
25.37 dB correct-token PSNR
18.91 dB wrong-token PSNR
6.46 dB gap
```

Also analyze new attack surfaces created by the retrieval database.

Steganalysis

If feasible, compare normal/reference images with stego images using:

- XuNet,
- SRNet.

Possible metrics:

- detection accuracy,
- AUC,
- false-positive rate,
- false-negative rate.

Lower detectability is preferable.

---

16. Required Ablations

Retrieval

```
CLIP semantic retrieval
vs
CLIP + structural reranking
```

Token

```
without token selection
vs
with token selection
```

VLM

```
without VLM
vs
VLM reranking
```

Editing

```
retrieval only
vs
retrieval + editing
```

Fallback

```
always retrieve
vs
retrieve-or-generate
```

Candidate pool

Vary Top-K and measure:

```
K → retrieval quality
  → reconstruction quality
  → security
  → latency
```

The exact K values should be selected after the retrieval prototype works; candidate-pool size affects both performance and the token-security interpretation.

---

17. Fair Experimental Matrix

For every test image:

```
Reference method
 ├─ Generation
 ├─ Retrieval
 └─ Retrieval + Editing
          ↓
      Same EDICT
          ↓
      Same channel
          ↓
      Same recovery
          ↓
PSNR / SSIM / MSE / LPIPS
+ latency / memory / security
```

Repeat across:

```
image class × SNR × reference method × token condition
```

Use the same:

- dataset,
- secret images,
- preprocessing,
- Kpub extraction,
- Xfeat extraction,
- EDICT configuration,
- token perturbation,
- semantic communication,
- channel,

- evaluation code.

---

18. Reproducibility Constraint

Transmitter and receiver must share or reproduce exactly:

```
Database version
Embedding model/checkpoint
Preprocessing
FAISS index
Ranking function/weights
Top-K
Token mapping
Editing parameters
EDICT parameters
```

Required invariant:

```
Xref_transmitter == Xref_receiver
```

If the references differ, EDICT recovery can fail.

Reference selection should therefore be deterministic even when an intelligent VLM is used for candidate scoring.

---

19. Project Structure

```
RACS/
├── data/
│   ├── secret_images/
│   ├── retrieval_images/
│   └── metadata/
├── semantic/
│   ├── semantic_extraction/
│   └── structure_extraction/
├── retrieval/
│   ├── embedding/
│   ├── faiss_index/
│   ├── ranking/
│   └── token_selection/
├── reference/
│   ├── retrieval/
│   └── lightweight_editing/
├── steganography/
│   ├── vae/
│   ├── edict/
│   └── token_perturbation/
├── communication/
│   ├── awgn/
│   ├── swinjscc/
│   └── digital_semcom/
├── security/
│   ├── wrong_token/
│   ├── no_token/
│   └── steganalysis/
├── evaluation/
│   ├── psnr.py
│   ├── ssim.py
│   ├── mse.py
│   ├── lpips.py
│   ├── latency.py
│   └── memory.py
└── experiments/
    ├── baseline/
    ├── retrieval/
    ├── ablation/
    └── results/
```

20. Risks and Mitigations

Risk	Consequence	Mitigation
Semantic retrieval is structurally poor	Lower EDICT recovery quality	Structural reranking, Top-K, editing
Token selection harms quality	Bad Xref	Select only from high-quality candidate pool
Editing is too expensive	Efficiency advantage disappears	Benchmark against full generation
Tx/Rx reference mismatch	Recovery failure	Deterministic shared configuration
Retrieval database creates attack surface	New security weakness	Avoid explicit IDs where possible; analyze attacker knowledge
Retrieval-only fails on unusual images	Large quality drop	Retrieve-or-generate fallback

---

21. Potential Research Contributions

Only claim contributions actually implemented and validated.

1. Retrieval-based reference-image acquisition for coverless semantic steganography.
2. Semantic + structural multimodal reference reranking.
3. Token-controlled deterministic reference selection.
4. Adaptive retrieve-or-generate reference acquisition.
5. Comprehensive quality-security-efficiency comparison between retrieval and diffusion generation.

---

22. What Counts as a Successful Result?

RACS does not have to beat AgentSemSteCom on every metric.

A scientifically useful result may be:

```
PSNR      slightly lower
SSIM      similar
LPIPS     similar
Security  similar
Latency   substantially lower
Memory    lower
```

Another useful result may be:

```
Retrieval-only:
fast but quality drops

Retrieval + editing:
moderate cost and recovered quality

Hybrid:
retrieval for easy cases,
generation for difficult cases
```

---

23. Negative Results Are Valid

Example:

```
Retrieval = 8× faster
but
PSNR decreases strongly
SSIM decreases
EDICT recovery becomes unstable
```

This supports the conclusion that pure semantic retrieval is insufficient for compatible reversible reference conditioning.

That can motivate:

```
structure-aware retrieval
→ light editing
→ retrieve-or-generate
```

The goal is not to force a positive result; it is to identify the quality-efficiency trade-off and its causes.

---

24. Final Research Questions

The completed project must answer:

Quality

- Does retrieval preserve reconstruction quality?
- How much do PSNR, SSIM, MSE and LPIPS change?
- Which image classes benefit/fail?

Efficiency

- How much reference latency is saved?
- How much GPU memory is saved?
- How much computation is saved?
- What is the cost of VLM reranking?
- What is the cost of editing?

Security

- Does the correct token enable recovery?
- Does a wrong token fail?
- What happens without a token?
- Does the retrieval database introduce new attack surfaces?

Robustness

- How does performance change from 5–20 dB?
- Which classes are most affected?

Retrieval

- How does Top-K affect quality?
- Does structural reranking help?
- Does VLM reranking justify its cost?
- Does token selection preserve quality?

Adaptive architecture

- When should the system retrieve?
- When should it edit?

- When should it fall back to generation?

---

25. Final Result Tables

Quality

System	SNR	PSNR ↑	SSIM ↑	MSE ↓	LPIPS ↓
AgentSemSteCom					
RACS Retrieval					
RACS + Editing					
RACS Hybrid					

Efficiency

System	Reference Time	Total Time	GPU Memory	Computation
AgentSemSteCom				
RACS Retrieval				
RACS + Editing				
RACS Hybrid				

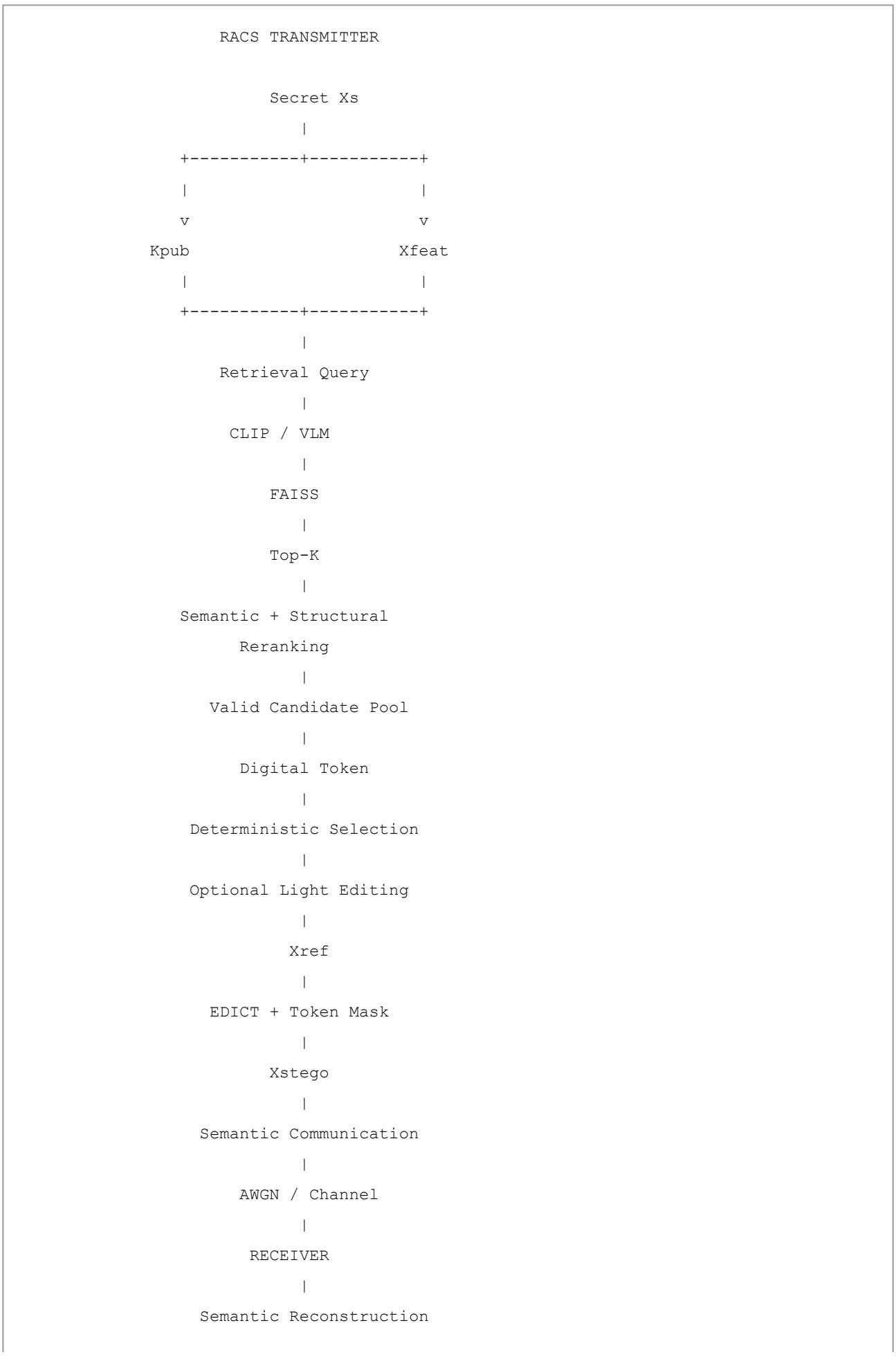
Security

System	Correct Token	Wrong Token	Gap
AgentSemSteCom			
RACS			

Do not invent values before experiments.

---

26. Final Architecture in One View


RACS TRANSMITTER
Secret Xs
Kpub
Xfeat
Retrieval Query
CLIP / VLM
FAISS
Top-K
Semantic + Structural
Reranking
Valid Candidate Pool
Digital Token
Deterministic Selection
Optional Light Editing
Xref
EDICT + Token Mask
Xstego
Semantic Communication
AWGN / Channel
RECEIVER
Semantic Reconstruction

```
                    |
          Same Retrieval Procedure
                    |
               Same Xref
                    |
            EDICT Inversion
                    |
         Inverse Token Mask
                    |
           Recovered Secret
```

---

27. Final Mental Model

Existing system

```
SECRET
 ↓
SEMANTIC + STRUCTURE
 ↓
GENERATE REFERENCE
 ↓
EDICT
 ↓
SEMANTIC COMMUNICATION
 ↓
CHANNEL
 ↓
EDICT RECOVERY
 ↓
SECRET
```

RACS

```
SECRET
 ↓
SEMANTIC + STRUCTURE
 ↓
RETRIEVE
 ↓
RERANK
 ↓
TOKEN SELECT
 ↓
OPTIONAL EDIT
 ↓
EDICT
 ↓
SEMANTIC COMMUNICATION
 ↓
CHANNEL
 ↓
RETRIEVE SAME REFERENCE
 ↓
EDICT RECOVERY
 ↓
SECRET
```

One-sentence definition

> ***RACS is a retrieval-augmented coverless semantic steganography framework that replaces the computationally expensive diffusion-based reference-image generation stage of AgentSemSteCom with deterministic multimodal retrieval and optional lightweight structural editing, while preserving the downstream EDICT-based hiding/recovery and semantic communication pipeline for a direct quality, security, robustness, and efficiency comparison.***

---

28. Immediate Implementation Direction

The next document is the **RACS Implementation Plan**. It must derive its phases directly from this specification.

The implementation should proceed from:

```
Environment
 ↓
VAE / latent understanding
 ↓
Diffusion understanding
 ↓
Retrieval database
 ↓
CLIP
 ↓
FAISS
 ↓
Token selection
 ↓
Structural retrieval
 ↓
EDICT
 ↓
RACS + EDICT
 ↓
AWGN / semantic communication
 ↓
VLM
 ↓
Light editing
 ↓
Retrieve-or-generate
 ↓
Security / steganalysis
 ↓
Ablations / final experiments
```

The first scientific version must remain:

```
same downstream pipeline
+
different reference acquisition
```

so every measured improvement or degradation can be attributed to RACS.